

Mathematical notes

Ancillary material for “The Last Human Gate”

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These notes derive the algebraic statements used in the paper. Results are referred to by name, as in the manuscript. Nothing here is estimated from data.

1 Workload ratio and break-even thresholds

With common visits λ and useful hours K , the labor account gives

$$\rho = \frac{N_A}{N_H} = \frac{\lambda \bar{h}_A + B_A}{\lambda h_0 + B_H} = \frac{\bar{h}_A/h_0 + b_A}{1 + b_H}, \quad b_s = \frac{B_s}{\lambda h_0}.$$

Write $h_E = \kappa \eta h_0$, $h_R = \mu h_S^H = \mu \sigma h_0 = m h_0$ and $h_W = r h_0$. Then $\bar{h}_A/h_0 = q\kappa\eta + (1-q)\sigma\mu + r$. The baseline partition states $q\kappa + (1-q)\sigma = 1$, so $(1-q)\sigma = 1 - \phi$ with $\phi = q\kappa$. Substituting gives both forms of the workload ratio:

$$\rho = \frac{q\kappa\eta + (1-q)\sigma\mu + r + b_A}{1 + b_H} = \frac{\phi\eta + (1-\phi)\mu + r + b_A}{1 + b_H}.$$

Exception-share threshold. In overall-mean units, $\rho < 1$ is $qk + (1-q)m + r + b_A < 1 + b_H$, that is $q(k - m) < 1 + b_H - m - r - b_A$. For $k > m$ this is the exception-share threshold $q < q^*$; for $k < m$ the inequality reverses; for $k = m$ the case share drops out. Solving the same equality for k at a given $q > 0$ gives $k^* = m + (1 + b_H - m - r - b_A)/q$, the line drawn in the frontier figure, once per route.

Exception-effort margin. In group-relative units, $\rho < 1$ is $\phi\eta < 1 + b_H - (1 - \phi)\mu - r - b_A$. For $\phi > 0$,

$$\eta^* = \frac{1 + b_H - (1 - \phi)\mu - r - b_A}{\phi} = \mu + \frac{1 + b_H - \mu - r - b_A}{\phi}, \quad \frac{\partial \eta^*}{\partial \phi} = -\frac{1 + b_H - \mu - r - b_A}{\phi^2}.$$

The margin therefore falls with ϕ exactly when $\mu + r + b_A < 1 + b_H$, that is, when a process made only of standard cases would save labor. In the synthetic example m is held common, so $(1 - \phi)\mu = (1 - q)m$ and the code evaluates $\eta^* = [1 + b_H - (1 - q)m - r - b_A]/\phi$.

2 Selection identity

Let $T \geq 0$ be baseline handling time with $\mathbb{E}[T] = h_0 > 0$, $J \in \{0, 1\}$ the residual indicator, $e(T) = \Pr(J = 1 \mid T)$ and $q = \mathbb{E}[e(T)] > 0$. Iterated expectations give $\mathbb{E}[TJ] = \mathbb{E}[Te(T)] = \text{Cov}(T, e(T)) + h_0 q$. Dividing by q yields $\mathbb{E}[T \mid J = 1] - h_0 = \text{Cov}(T, e(T))/q$; dividing by h_0 yields $\phi = q + \text{Cov}(T, e(T))/h_0$. Since $\kappa = \mathbb{E}[T \mid J = 1]/h_0$, also $\phi = q\kappa$. For independent copies T, T' , $2 \text{Cov}(T, e(T)) = \mathbb{E}[(T - T')(e(T) - e(T'))] \geq 0$ when e is nondecreasing, because the two factors then share a sign. The identities restrict first moments and one cross-moment only: truncating a distribution to a narrow band of difficult cases raises the conditional mean and can lower the conditional variance.

3 Strategic gate model

Setup. Let ε have distribution function F and density f . With $s = \xi + u + g$, approval probability is $p(s) = \Pr(s + \varepsilon \geq \tau) = 1 - F(\tau - s)$, so $p'(s) = f(\tau - s) \geq 0$ and $p''(s) = -f'(\tau - s)$. The project maximizes $\Pi(u, g) = Vp(s) - c_u(u) - c_g(g) - D(g, a)$ over $u, g \geq 0$.

First- and second-order conditions. At an interior optimum,

$$Vp'(s) - c'_u(u) = 0, \quad Vp'(s) - c'_g(g) - D_g(g, a) = 0.$$

Put $t = Vp''(s)$, $\omega_u = c''_u(u) > 0$ and $\omega_g = c''_g(g) + D_{gg}(g, a) > 0$. The Hessian is

$$H = \begin{pmatrix} t - \omega_u & t \\ t & t - \omega_g \end{pmatrix}, \quad \det H = \omega_u \omega_g - t(\omega_u + \omega_g) = \Delta.$$

A negative-definite Hessian is characterized by $t - \omega_u < 0$ and $\Delta > 0$. These sufficient second-order conditions for a strict local maximum hold automatically when $t \leq 0$.

Assurance intensity. Differentiating the two conditions with respect to a ,

$$H \begin{pmatrix} du \\ dg \end{pmatrix} = \begin{pmatrix} 0 \\ D_{ga} da \end{pmatrix} \implies \frac{\partial u^*}{\partial a} = -\frac{t D_{ga}}{\Delta}, \quad \frac{\partial g^*}{\partial a} = \frac{(t - \omega_u) D_{ga}}{\Delta}, \quad \frac{\partial s^*}{\partial a} = -\frac{\omega_u D_{ga}}{\Delta}.$$

With $D_{ga} > 0$: cosmetic effort falls whenever the second-order conditions hold; the observable score falls as well; genuine remediation rises weakly if and only if $t \leq 0$. If $t > 0$ (and the second-order conditions hold), both efforts fall.

Lemma 1. *If f is differentiable and unimodal with mode zero, then $p''(s) \leq 0$ whenever $s \geq \tau$. The converse also holds if $f'(x) < 0$ for every $x > 0$.*

Proof. Unimodality with mode zero means $f'(x) \geq 0$ for $x \leq 0$ and $f'(x) \leq 0$ for $x \geq 0$. Thus $s \geq \tau$ gives $\tau - s \leq 0$ and $p''(s) = -f'(\tau - s) \leq 0$. Under the additional strict derivative condition, $s < \tau$ gives $p''(s) > 0$, proving the converse. Without that condition, a flat density segment, including a zero-density tail, can give $p''(s) = 0$ even below the threshold. $\square$

The redirection result therefore applies to projects whose expected score already meets the threshold; below it, stronger assurance can discourage both kinds of effort.

Corner solutions. If $g^* = 0$ because $Vp'(s) \leq c'_g(0) + D_g(0, a)$, a marginal increase in a leaves (u^*, g^*) unchanged. If $u^* = 0$ is binding and $g^* > 0$, the single remaining condition gives $\partial g^* / \partial a = D_{ga} / (t - \omega_g) < 0$ under its second-order condition $t - \omega_g < 0$.

Clearer criteria. Let a clarity parameter θ lower both marginal costs: $c'_{u\theta} < 0$ and $c'_{g\theta} < 0$. Then

$$\frac{\partial u^*}{\partial \theta} = \frac{c'_{u\theta}(t - \omega_g) - t c'_{g\theta}}{\Delta}, \quad \frac{\partial g^*}{\partial \theta} = \frac{(t - \omega_u) c'_{g\theta} - t c'_{u\theta}}{\Delta}.$$

For $t \leq 0$ the first term of each numerator is positive and the second is nonpositive, so neither sign is determined: clearer criteria can raise genuine work, cosmetic work, or both.

Numerical check. `test_reproduce.py` solves the model with quadratic costs, logistic noise and $D(g, a) = ag^2/2$ at two assurance levels and confirms the predicted signs in the region $s > \tau$.

4 Reliability bounds and the readiness example

For events $\mathcal{C}_1, \dots, \mathcal{C}_n$ with $\Pr(\mathcal{C}_i) = p_i$, the Fréchet inequalities give $\max(0, 1 - \sum_i (1 - p_i)) \leq \Pr(\bigcap_i \mathcal{C}_i) \leq \min_i p_i$. With $n = 20$ and $p_i = 0.99$ the bounds are 0.80 and 0.99; independence gives $0.99^{20} \approx 0.8179$.

The geometric readiness index of $(0.01, 1, 1, 1, 1)$ is $0.01^{1/5} = 10^{-0.4} \approx 0.398$, which shows that the index is only partially sensitive to a near-zero dimension and motivates reporting a minimum index and hard blockers alongside it.